Problem Statement

Project: E-Commerce Sales & Customer Analytics

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Tools: PostgreSQL, Power BI, DAX

1. Background

An online retailer sells through four sales channels (Mobile App, Website, Marketplace and Social Media) to customers in seven countries. Between January 2021 and December 2025 it recorded 138,116 orders from 24,911 customers.

Management has the sales numbers but not a clear picture of how the business is really performing. Yearly revenue looks level, and there is no breakdown of where profit comes from, where it is lost, or what is affecting customer satisfaction.

2. Problem

The company needs to know whether the business is growing, where revenue and profit are being lost, and which customers, products, discounts and operations are helping or hurting results, so that it can decide what to change.

3. Questions this project answers

1. Is the data complete and free of duplicates, and what period does it cover?
2. What are the overall revenue, profit, margin, customer count and average order value?
3. Is revenue growing from year to year?
4. What share of orders is cancelled, returned or left pending, and how much revenue does that represent?
5. How does revenue change month to month, and is there a seasonal pattern?
6. Which sales channel performs best?
7. Which product categories bring in the most revenue, and which earn the best margin?
8. Which brands earn the most profit, and at what margin?
9. Do heavier discounts reduce profit?
10. How do customer segments differ in order value, margin and discount?
11. Which countries bring in the most revenue?
12. Which marketing channels bring in the most revenue?
13. How much revenue comes from repeat customers?
14. How often do payments fail, and does it depend on the payment method?
15. How often are orders delayed, and does it depend on the shipping method?
16. Does late delivery lower customer ratings?
17. Why do customers return products, and are the reasons avoidable?
18. Based on these answers, what should the business do first?

4. Data used

Table	Rows	Description
orders	138,116	One row per order: status, dates, channels, payment, delivery, rating, net sales, profit
order_items	397,569	One row per product in an order: quantity, discount, net sales, profit
product_catalog	1,175	Product, category, brand and cost
customer_master	25,000	Customer segment, country, age and acquisition cost

5. Key metrics

Metric	Definition
Revenue	Sum of `net_sales`: the amount charged after discount, including tax and shipping
Profit	`net_sales` minus product cost and shipping cost
Profit margin	Profit divided by revenue
Average order value	Revenue divided by number of orders
Unrealised revenue	Revenue from cancelled, returned and pending orders
Discount band	Discount amount divided by gross sales, grouped into under 10%, 10-20%, 20-30%, 30-40% and 40% and above
Late delivery rate	Share of completed orders with delivery status "Delayed"
Repeat-customer group	Customers grouped by number of orders: 1, 2-3, 4-6 and 7+

6. Approach

1. **Check the data** in PostgreSQL: table sizes, duplicates, missing values and date range.
2. **Measure the business**: headline KPIs and the split of order outcomes.
3. **Analyse in SQL**: time, channel, product, discount, customer, marketing, payment, delivery and returns.
4. **Build the dashboard** in Power BI: four pages on a linked data model with DAX measures.
5. **Write the report**: findings for each question and a prioritised list of recommendations.

7. Deliverables

- Ecommerce_SQL_Analysis.sql
- Power BI Dashboard: Business Overview, Sales Trends, Customers & Marketing, and Category & Discount Analysis
- Ecommerce_Insights_Report.pdf